How Moirai-MoE makes a forecast

Salesforce · 117M parameters, 11M used per token (small) · decoder-only mixture of experts · zero-shot

1. All rows in one sequence

Targets and covariates are laid end to end as one long sequence. A ? is a value not known yet.

past *future*

Target temperature

Known ahead day length

Past only NAO index

target **known** **past**

joined into one sequence

Any number of rows fits, so the five metals share one sequence.

2. Scale, then patches of 16

Each row is scaled by the mean μ and standard deviation σ of its own past. One patch size for every frequency.

$$\tilde{x} = \frac{x - \mu}{\sigma}$$

value

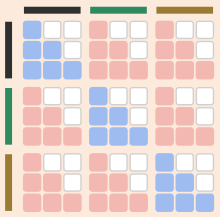
one patch = 16 steps

residual MLP

one token: 384 numbers

3. Six causal blocks with experts

Each token reads the tokens up to its own time, in every row. RMSNorm in front, shortcut around.



same row other row

Attention: RoPE in time, plus a same-row / other-row bias

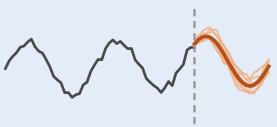
gate

2 of 32 experts run for this token

Experts: the gate scores each token against 32 cluster centres from a pre-trained Moirai; each expert is a SwiGLU feed forward (512)

4. Sample patch by patch

Each token gives a **mix of four distributions** for every step of the next patch. A sample is drawn, appended, and the model runs again.



median

sample paths

100 paths; their median is the point forecast.

Experts specialise on kinds of tokens, so one patch size serves every frequency.

6 blocks · width 384 · 6 heads · 32 experts, 2 per token · patches of 16 · mixture: Student-t, normal, negative binomial, log-normal

In the notebooks: last 2,048 days or full monthly history · all covariates · five metals together · 100 samples · 90 days = 6 rounds